Status	Finished
Started	Thursday, 20 August 2026, 3:38 PM
Completed	Thursday, 20 August 2026, 3:52 PM
Duration	13 mins 37 secs
Grade	**13.67** out of 16.00 (**85.42**%)

Question 1

Complete

Mark 0.67 out of 1.00

Match the following filters to the appropriate description

1. A filter used to remove high-frequency noise from a signal. Low-pass filter
2. A filter that allows only signals within a specific frequency range (f_1 to f_2) to pass. Band-pass filter
3. A filter commonly used to select a desired frequency band in analog transmission. Frequency-pass filter

The correct answer is:
Match the following filters to the appropriate description

1. A filter used to remove high-frequency noise from a signal. [Low-pass filter]
2. A filter that allows only signals within a specific frequency range (f_1 to f_2) to pass. [Band-pass filter]
3. A filter commonly used to select a desired frequency band in analog transmission. [Band-pass filter]

Question 2

Complete

Mark 0.00 out of 2.00

A signal has an initial power of 10 mW. After passing through a transmission medium, its power decreases to 0.5 mW. Calculate the power change in decibels (dB).

Assume $\log_{10}(2) \approx 0.3$ and $\log_{10}(5) \approx 0.7$.

Answer: 100 dB

Gain/Loss (dB)=$10\log_{10}(P1/P2)$

where P_1=10 mW, P_2=0.5 mW

$10\log_{10}(0.5/10) = 10\log_{10}(5/100) = 10(\log_{10}5-\log_{10}100) = 10(0.7 -2) = 7 - 20 = -13$ dB

Question 3

Complete

Mark 1.00 out of 1.00

Find an appropriate terminology for each statement.

- The signal becomes weaker as it travels through the transmission medium because its power decreases. Attenuation
- Unwanted signals from the environment interfere with the transmitted signal and may corrupt the received data. Noise
- When different frequency components experience different delays, the shape of the received signal may change. Distortion

The correct answer is:
Find an appropriate terminology for each statement.

- The signal becomes weaker as it travels through the transmission medium because its power decreases. [Attenuation]
- Unwanted signals from the environment interfere with the transmitted signal and may corrupt the received data. [Noise]
- When different frequency components experience different delays, the shape of the received signal may change. [Distortion]

Question 4

Complete

Mark 1.00 out of 1.00

Convert Watts (W) to dBm and dBm to Watts (W).

- 1 W = [1000] mW = [30] dBm
- -20 dBm = [0.01] mW

0.01	0.001	100	1	0	10	30	1000	20

$P_{dBm}=10\log_{10}(P_{mW})$

Watt to dBm

1 W=1000 mW

$10\log_{10}(1000)=30$dBm

dBm to Watt

-20 dBm = $10\log_{10}(P_{mW})$ --> $P_{mW} = 10^{-20/10} = 10^{-2} = 0.01$ mW

The correct answer is:
Convert Watts (W) to dBm and dBm to Watts (W).

- 1 W = [1000] mW = [30] dBm
- -20 dBm = [0.01] mW

Question 5

Complete

Mark 2.00 out of 2.00

A communication channel has a bandwidth of 5000 Hz and uses 8 distinct signal levels. Assume that noise can be neglected.

Which theorem should be used to determine the maximum bit rate? Nyquist

What is the maximum bit rate in kbps? 30

$R_{max}=2B\log_2 L$

when B=5000 Hz, L=8 and $\log_2 8=3$

therefore $R_{max}=2(5000)(3)=30000$ bps=30 kbps

The correct answer is:
A communication channel has a bandwidth of 5000 Hz and uses 8 distinct signal levels. Assume that noise can be neglected.

Which theorem should be used to determine the maximum bit rate? [Nyquist]

What is the maximum bit rate in kbps? [30]

Question 6

Complete

Mark 2.00 out of 2.00

Consider a noisy channel with a bandwidth of 10 kHz and a signal-to-noise ratio (SNR) of 15.

Which theorem should be used to determine the maximum bit rate? Shannon

What is the maximum bit rate in kbps? 40

$C=B\log_2(1+SNR)$

$C=10{,}000\log_2(1+15) = 10{,}000\log_2(16) = 10{,}000(4)=40{,}000$ bps

The correct answer is:
Consider a noisy channel with a bandwidth of 10 kHz and a signal-to-noise ratio (SNR) of 15.

Which theorem should be used to determine the maximum bit rate? [Shannon]

What is the maximum bit rate in kbps? [40]

Question 7

Complete

Mark 2.00 out of 2.00

A network carries 6000 frames per minute, with 10,000 bits in each frame. What is the throughput?

Answer: 1 Mbps

$6000*10000/60 = 1*10^6 = 1$ Mbps

Question 8

Complete

Mark 2.00 out of 2.00

A signal travels 2400 km through a medium at 2.4×10^8m/s.

The propagation time = 10 ms

Tp = $(2.4*10^6)/(2.4*10^8)$=0.01s = 10ms

Question 9

Complete

Mark 1.00 out of 1.00

Which three factors are used in the lecture when considering the achievable data rate of a channel?

a. Cable color, connector type, and computer model

b. Bandwidth, signal levels, and channel quality

c. Sampling phase, packet route, and application protocol

d. MAC address, IP address, and port number

e. Frame size, packet size, and application size

The correct answer is: Bandwidth, signal levels, and channel quality

Question 10

Complete

Mark 2.00 out of 2.00

A signal enters a medium at 8 mW. After passing through, the power is measured at 2 mW.

What is the attenuation in decibels? (log 2 ≈ 0.3)

Answer: -6 dB.

Gain/Loss (dB)=$10\log_{10}(P2/P1)$

where P1=8 mW,P2=2 mW

therefore $10\log_{10}(2/8)=10\log_{10}(1/4)=10\log_{10}(2^{-2})=10(-2)(0.3)$ = -6